

Week 0 Example: Cancer Data

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Loading Packages

The summarytools package provides dfSummary() for data summaries.

```
# summarytools provides dfSummary() for data summaries  
library(summarytools)
```

Importing Data

The data file lives on the course GitHub site. Store the web address in an object, then hand that address to read.csv().

```
# store the web address of the raw data file  
filepath <-  
  ↪ "https://raw.githubusercontent.com/craigenders/psych250a/main/data/Cancer.csv"  
  
# read the file at that address into a data frame named Cancer  
Cancer <- read.csv(filepath, stringsAsFactors = TRUE)
```

Converting Categorical Variables to Factors

Diagnosis and Male are stored as 0/1 codes. factor() converts each one to a categorical variable with descriptive labels.

```
# convert Diagnosis from 0/1 codes to a factor with descriptive labels
Cancer$Diagnosis <- factor(Cancer$Diagnosis,
                           levels = c(0, 1),
                           labels = c("Non-malignant", "Malignant"))

# convert Male from 0/1 codes to a factor with descriptive labels
Cancer$Male <- factor(Cancer$Male,
                     levels = c(0, 1),
                     labels = c("Female", "Male"))

# print the first few rows to confirm the labels replaced the codes
head(Cancer)
```

	Participant	Diagnosis	Age	Male	Comorbids	Optimism	Depression	VisImpair
1	1	Non-malignant	43	Female	1	7	42	52
2	2	Non-malignant	25	Female	1	6	3	43
3	3	Non-malignant	55	Female	0	10	8	61
4	4	Non-malignant	46	Male	0	8	22	50
5	5	Non-malignant	73	Male	2	12	3	24
6	6	Malignant	60	Female	1	12	10	30

Summarizing Data

dfSummary() gives an overview of every variable in the data frame.

```
# overview of every variable in the data frame
dfSummary(Cancer)
```

Data Frame Summary
 Cancer
 Dimensions: 299 x 8
 Duplicates: 0

No	Variable	Stats / Values	Freqs (% of Valid)	Graph
↪ Valid	Missing			
1	Participant	Mean (sd) : 150 (86.5)	299 distinct values	: : : : : :
↪ 299	0			
	[integer]	min < med < max:	(Integer sequence)	: : : : : :
	↪ (100.0%)	(0.0%)		
		1 < 150 < 299		: : : : : :
		IQR (CV) : 149 (0.6)		: : : : : :
				: : : : : :
2	Diagnosis	1. Non-malignant	192 (64.2%)	IIIIIIIIIIII
↪ 299	0			
	[factor]	2. Malignant	107 (35.8%)	IIIIIII
	↪ (100.0%)	(0.0%)		

```

3   Age           Mean (sd) : 59 (13.9)      61 distinct values      . :
  ↪ 299           0
    [integer]     min < med < max:           : :
      ↪ (100.0%)   (0.0%)
                    19 < 60 < 91             . : : .
                    IQR (CV) : 17.5 (0.2)     : : : :
                                           . : : : : :

4   Male          1. Female          138 (46.2%)      I I I I I I I I
  ↪ 299           0
    [factor]      2. Male            161 (53.8%)      I I I I I I I I I
      ↪ (100.0%)   (0.0%)

5   Comorbids     Mean (sd) : 0.9 (0.7)      0 : 91 (30.4%)      I I I I I
  ↪ 299           0
    [integer]     min < med < max:           1 : 135 (45.2%)  I I I I I I I I I
      ↪ (100.0%)   (0.0%)
                    0 < 1 < 2              2 : 73 (24.4%)      I I I I
                    IQR (CV) : 1 (0.8)

6   Optimism      Mean (sd) : 8.6 (2.9)      13 distinct values      :
  ↪ 299           0
    [integer]     min < med < max:           :
      ↪ (100.0%)   (0.0%)
                    0 < 9 < 12             :
                    IQR (CV) : 4 (0.3)       : . : : :
                                           . . . . : : : : :

7   Depression    Mean (sd) : 14.9 (11.4)    50 distinct values      :
  ↪ 299           0
    [integer]     min < med < max:           :
      ↪ (100.0%)   (0.0%)
                    0 < 11 < 60             : : .
                    IQR (CV) : 13 (0.8)     : : : .
                                           : : : : . . . .

8   VisImpair     Mean (sd) : 42 (15.1)      66 distinct values      :
  ↪ 299           0
    [integer]     min < med < max:           . : :
      ↪ (100.0%)   (0.0%)
                    10 < 40 < 90            : : : :
                    IQR (CV) : 20.5 (0.4)   : : : : .
                                           : : : : : :

```

```

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↪ -----

```